

RailwaySentimentAnalysis

March 15, 2026

1 Railway Sentiment Analysis

```
[5]: import nltk
nltk.download('punkt_tab') #tokenizer
```

```
[nltk_data] Downloading package punkt_tab to
[nltk_data] /Users/kishankunal/nltk_data...
[nltk_data] Unzipping tokenizers/punkt_tab.zip.
```

[5]: True

```
[14]: import pandas as pd
df = pd.read_csv("./data/train.csv", encoding = "latin-1", header = None,
    ↪names = ['index', 'sentiment_label', 'text']) #kept data in same directory
    ↪under folder name data
df.head()
```

```
[14]:
```

	index	sentiment_label	text
0	Item ID	Sentiment	SentimentText
1	1	0	@RailMinIndia My PNR is 8348062961, I am in wa...
2	2	0	@sureshprabhu @RailMinIndia AC not working in...
3	3	0	@RailMinIndia I'm traveling to chennai by trai...
4	4	5	@RailMinIndia irtc is not responding at the t...

```
[15]: df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1367 entries, 0 to 1366
Data columns (total 3 columns):
#   Column          Non-Null Count  Dtype
---  -
0   index           1367 non-null   object
1   sentiment_label  1367 non-null   object
2   text            1367 non-null   object
dtypes: object(3)
memory usage: 32.2+ KB
```

```
[16]: df.drop(0, inplace=True) #removing header as it is redundant
```

```
[19]: #Displaying schema and null check
print(df.info())
print(df.isnull().sum())
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1366 entries, 1 to 1366
Data columns (total 3 columns):
#   Column          Non-Null Count  Dtype
---  -
0   index           1366 non-null   object
1   sentiment_label  1366 non-null   object
2   text            1366 non-null   object
dtypes: object(3)
memory usage: 32.1+ KB
None
index           0
sentiment_label 0
text            0
dtype: int64
```

```
[26]: import matplotlib.pyplot as plt
import seaborn as sns

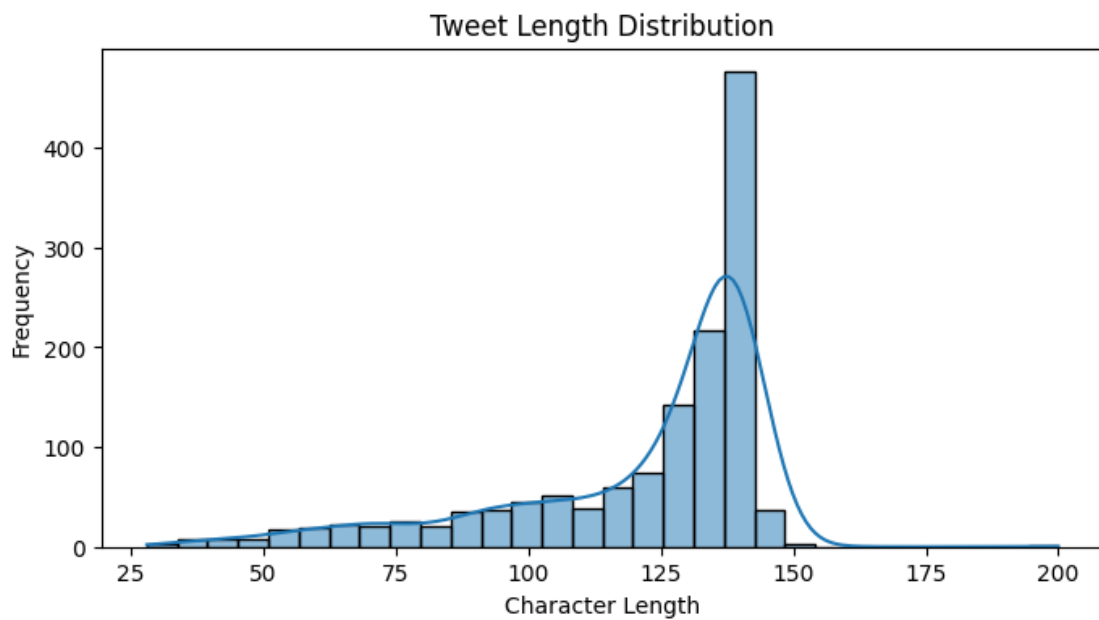
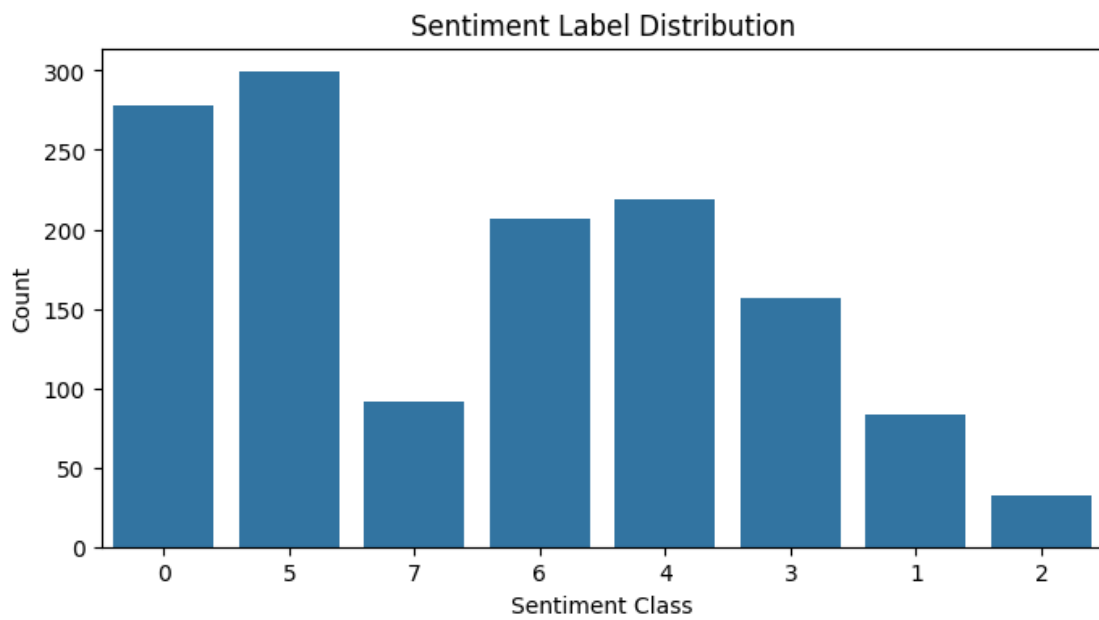
# describe dataset
print(df.describe())

# Distribution of sentiment labels
plt.figure(figsize=(8,4))
sns.countplot(x='sentiment_label', data=df)
plt.title('Sentiment Label Distribution')
plt.xlabel('Sentiment Class')
plt.ylabel('Count')
plt.show()

# Tweet Length Distribution
df['tweet_length'] = df['text'].apply(len)
plt.figure(figsize=(8,4))
sns.histplot(df['tweet_length'], bins = 30, kde=True)
plt.title('Tweet Length Distribution')
plt.xlabel('Character Length')
plt.ylabel('Frequency')
plt.show()
```

```
      index sentiment_label \
count   1366             1366
unique   1366              8
top         1              5
freq         1             299
```

count	1366
unique	1353
top	@rpfwrbct @rpfwr1 @RailMinIndia @Deshbhakth1 @...
freq	2



- Sentiment labels range from 0 (Negative) to 7(Positive)
- A few outliers with unusually long lengths (~200+) likely due to urls or concatenated message

2 1.User-Mention Extraction

```
[28]: import re #regular expression library

def extract_mentions(text):
    return re.findall(r'(@\w+)', text) #checking user handle @

df['mentions'] = df['text'].apply(extract_mentions)
```

```
[29]: df.head()
```

```
[29]:
```

	index	sentiment_label	text \
1	1	0	@RailMinIndia My PNR is 8348062961, I am in wa...
2	2	0	@sureshprabhu @RailMinIndia AC not working in...
3	3	0	@RailMinIndia I'm traveling to chennai by trai...
4	4	5	@RailMinIndia irtc is not responding at the t...
5	5	7	@DRMbhopal @RailMinIndia @sanjaygupta2012 @drm...

	tweet_length	mentions
1	140	[@RailMinIndia]
2	137	[@sureshprabhu, @RailMinIndia]
3	122	[@RailMinIndia]
4	68	[@RailMinIndia]
5	104	[@DRMbhopal, @RailMinIndia, @sanjaygupta2012, ...]

3 2. Hashtag extraction

```
[35]: def extract_hashtags(text):
    return [tag.lower() for tag in re.findall(r'#(\w+)', text)]

df['hashtags'] = df['text'].apply(extract_hashtags)
df['hashtags'] = df['hashtags'].apply(lambda x: list(set(x)))
df.head(20)
```

```
[35]:
```

	index	sentiment_label	text \
1	1	0	@RailMinIndia My PNR is 8348062961, I am in wa...
2	2	0	@sureshprabhu @RailMinIndia AC not working in...
3	3	0	@RailMinIndia I'm traveling to chennai by trai...
4	4	5	@RailMinIndia irtc is not responding at the t...
5	5	7	@DRMbhopal @RailMinIndia @sanjaygupta2012 @drm...
6	6	6	@RailMinIndia If you can't give justice to gra...
7	7	4	@sureshprabhu @RailMinIndiaPlz wrkout smthng ...
8	8	0	@RailMinIndia @mumbairailusers Dirty water flo...

9	9	6	Enter to win \$150 Amazon Gift Card! #Books #PN...
10	10	3	@RailMinIndia still vendors are selling local ...
11	11	4	@Central_Railway @RailMinIndia @drmmumbaicr pl...
12	12	0	@RailMinIndia #Swatch Bharat Abhiyaan# Train s...
13	13	5	@RailMinIndia @sureshpprabhu my reserved ticke...
14	14	4	@RailMinIndia update:- today intercity 12930 ...
15	15	0	@RailMinIndia TRAIN NO 12656 THERE IS NO PROP...
16	16	6	The Google il button is used 5 billion times p...
17	17	3	@RailMinIndia Sir, ?? ?????? ?????? ?????? ?? ?...
18	18	4	@RailMinIndia Train has departed Varanasi and ...
19	19	4	@sureshpprabhu @RailMinIndia student's pnr 21...
20	20	5	@RailMinIndia Dear Sir, my father is a senior ...

	tweet_length	mentions \
1	140	[@RailMinIndia]
2	137	[@sureshpprabhu, @RailMinIndia]
3	122	[@RailMinIndia]
4	68	[@RailMinIndia]
5	104	[@DRMbhopal, @RailMinIndia, @sanjaygupta2012, ...
6	127	[@RailMinIndia]
7	139	[@sureshpprabhu, @RailMinIndiaPlz]
8	131	[@RailMinIndia, @mumbairailusers]
9	94	[]
10	117	[@RailMinIndia]
11	120	[@Central_Railway, @RailMinIndia, @drmmumbaicr]
12	140	[@RailMinIndia]
13	137	[@RailMinIndia, @sureshpprabhu]
14	58	[@RailMinIndia]
15	121	[@RailMinIndia]
16	74	[]
17	140	[@RailMinIndia]
18	103	[@RailMinIndia]
19	140	[@sureshpprabhu, @RailMinIndia]
20	140	[@RailMinIndia]

	hashtags
1	[]
2	[]
3	[]
4	[]
5	[]
6	[]
7	[]
8	[]
9	[romance, books, urbanfantasy, pnr]
10	[]
11	[]

```

12             [no, swatch]
13             []
14             []
15             []
16             []
17             []
18             []
19             []
20             []

```

4 3.Extract urls

```

[40]: def extract_urls(text):
        urls = re.findall(r'https?://\S+', text)
        return urls

df['urls'] = df['text'].apply(extract_urls)
df['clean_text'] = df['text'].apply(lambda x: re.sub(r'https?://\S+', '', x))
df.head(20)

```

```

[40]:   index sentiment_label      text \
1      1              0  @RailMinIndia My PNR is 8348062961, I am in wa...
2      2              0  @sureshprabhu @RailMinIndia AC not working in...
3      3              0  @RailMinIndia I'm traveling to chennai by trai...
4      4              5  @RailMinIndia irtc is not responding at the t...
5      5              7  @DRMbhopal @RailMinIndia @sanjaygupta2012 @drm...
6      6              6  @RailMinIndia If you can't give justice to gra...
7      7              4  @sureshprabhu @RailMinIndiaPlz wrkout smthng ...
8      8              0  @RailMinIndia @mumbairailusers Dirty water flo...
9      9              6  Enter to win $150 Amazon Gift Card! #Books #PN...
10     10              3  @RailMinIndia still vendors are selling local ...
11     11              4  @Central_Railway @RailMinIndia @drmmumbaicr pl...
12     12              0  @RailMinIndia #Swatch Bharat Abhiyaan# Train s...
13     13              5  @RailMinIndia @sureshprabhu my reserved ticke...
14     14              4  @RailMinIndia  update:- today intercity 12930 ...
15     15              0  @RailMinIndia  TRAIN NO 12656 THERE IS NO PROP...
16     16              6  The Google il button is used 5 billion times p...
17     17              3  @RailMinIndia Sir, ?? ?????? ?????? ?????? ?? ?...
18     18              4  @RailMinIndia Train has departed Varanasi and ...
19     19              4  @sureshprabhu  @RailMinIndia student's pnr 21...
20     20              5  @RailMinIndia Dear Sir, my father is a senior ...

      tweet_length      mentions \
1              140      [@RailMinIndia]
2              137  [@sureshprabhu, @RailMinIndia]
3              122      [@RailMinIndia]

```

4	68	[@RailMinIndia]
5	104	[@DRMbhopal, @RailMinIndia, @sanjaygupta2012, ...]
6	127	[@RailMinIndia]
7	139	[@sureshprabhu, @RailMinIndiaPlz]
8	131	[@RailMinIndia, @mumbairailusers]
9	94	[]
10	117	[@RailMinIndia]
11	120	[@Central_Railway, @RailMinIndia, @drmmumbaicr]
12	140	[@RailMinIndia]
13	137	[@RailMinIndia, @sureshprabhu]
14	58	[@RailMinIndia]
15	121	[@RailMinIndia]
16	74	[]
17	140	[@RailMinIndia]
18	103	[@RailMinIndia]
19	140	[@sureshprabhu, @RailMinIndia]
20	140	[@RailMinIndia]

	hashtags	urls \
1	[]	[]
2	[]	[]
3	[]	[]
4	[]	[]
5	[]	[]
6	[]	[]
7	[]	[]
8	[]	[https://t.co/mBZUOFkLXQ]
9	[romance, books, urbanfantasy, pnr]	[https://t.co/jBfXzDe7P2]
10	[]	[https://t.co/wdFyLlmMek]
11	[]	[]
12	[no, swatch]	[https://t.co/h5uUdJuvel]
13	[]	[]
14	[]	[]
15	[]	[https://t.co/VIQmURi8HK]
16	[]	[]
17	[]	[https://t.co/sMfHUZdN2U]
18	[]	[]
19	[]	[]
20	[]	[]

	clean_text
1	@RailMinIndia My PNR is 8348062961, I am in wa...
2	@sureshprabhu @RailMinIndia AC not working in...
3	@RailMinIndia I'm traveling to chennai by trai...
4	@RailMinIndia irtc is not responding at the t...
5	@DRMbhopal @RailMinIndia @sanjaygupta2012 @drm...
6	@RailMinIndia If you can't give justice to gra...

```

7  @sureshprabhu @RailMinIndiaPlz wrkout smthng ...
8  @RailMinIndia @mumbairailusers Dirty water flo...
9  Enter to win $150 Amazon Gift Card! #Books #PN...
10 @RailMinIndia still vendors are selling local ...
11 @Central_Railway @RailMinIndia @drmmumbaicr pl...
12 @RailMinIndia #Swatch Bharat Abhiyaan# Train s...
13 @RailMinIndia @sureshprabhu my reserved ticke...
14 @RailMinIndia update:- today intercity 12930 ...
15 @RailMinIndia TRAIN NO 12656 THERE IS NO PROP...
16 The Google i1 button is used 5 billion times p...
17 @RailMinIndia Sir, ?? ?????? ?????? ?????? ?? ?...
18 @RailMinIndia Train has departed Varanasi and ...
19 @sureshprabhu @RailMinIndia student's pnr 21...
20 @RailMinIndia Dear Sir, my father is a senior ...

```

5 4. extracting emojis

```

[43]: import emoji

def replace_emoji(text):
    return emoji.demojize(text, delimiters=(" : ", ": "))

df['clean_text'] = df['clean_text'].apply(replace_emoji)
df['clean_text'] = df['clean_text'].str.replace(r':[a-zA-Z0-9_]+:', ' ',
regex=True)

df.head()

```

```

[43]:   index  sentiment_label      text \
1      1                0  @RailMinIndia My PNR is 8348062961, I am in wa...
2      2                0  @sureshprabhu @RailMinIndia AC not working in...
3      3                0  @RailMinIndia I'm traveling to chennai by trai...
4      4                5  @RailMinIndia irtct is not responding at the t...
5      5                7  @DRMbhopal @RailMinIndia @sanjaygupta2012 @drm...

      tweet_length      mentions hastags \
1          140      [@RailMinIndia]      []
2          137  [@sureshprabhu, @RailMinIndia]      []
3          122      [@RailMinIndia]      []
4           68      [@RailMinIndia]      []
5          104  [@DRMbhopal, @RailMinIndia, @sanjaygupta2012, ...]      []

      urls      clean_text
1      []  @RailMinIndia My PNR is 8348062961, I am in wa...
2      []  @sureshprabhu @RailMinIndia AC not working in...
3      []  @RailMinIndia I'm traveling to chennai by trai...

```



```

4  []  @RailMinIndia irctc is not responding at the t...
5  []  @DRMbhopal @RailMinIndia @sanjaygupta2012 @drm...

```

6 5. Text Pre-processing with NLTK & spaCy

```

[48]: from nltk.tokenize import word_tokenize
import spacy

nlp = spacy.load('en_core_web_sm')

#sample 100 random tweets

sample_tweets = df.sample(n=100)['clean_text']

nltk_tokens = sample_tweets.apply(word_tokenize)

spacy_tokens = sample_tweets.apply(lambda x: [token.text for token in nlp(x)])

#example comparison
for i in range(5):
    print(f"\nTweet {i+1} :")
    print("NLTK : ", nltk_tokens.iloc[i])
    print("SpaCy : ", spacy_tokens.iloc[i])

```

Tweet 1 :

```

NLTK :  ['@', 'RailMinIndia', 'Railway', 'was', 'never', 'been', 'so',
'irresponsible', '.', 'so', 'called', 'fake', 'development', 'is', 'happening',
'.', 'railway', 'was', 'never', 'been', 'so', 'irresponsible', '.', '?', '?',
'?', '?']

```

```

SpaCy :  ['@RailMinIndia', 'Railway', 'was', 'never', 'been', 'so',
'irresponsible', '.', 'so', 'called', 'fake', 'development', 'is', 'happening',
'.', 'railway', 'was', 'never', 'been', 'so', 'irresponsible', '.', '?', '?',
'?', '?']

```

Tweet 2 :

```

NLTK :  ['@', 'leomanojc', '@', 'SirDebasis', '@', 'Naren_Reddy23', '@',
'SayNoToHindi', '@', 'mnsadhikrut', '@', 'abpmajhatv', '@', 'anilshidore', '@',
'RailMinIndiaâ\x80']

```

```

SpaCy :  ['@leomanojc', '@SirDebasis', '@Naren_Reddy23', '@SayNoToHindi',
 '@mnsadhikrut', '@abpmajhatv', '@anilshidore', '@RailMinIndiaâ\x80', '']

```

Tweet 3 :

```

NLTK :  ['RT', '@', 'RailMinIndia', ':', 'Member', 'Traffic', 'reviewed', 'the',
'April-', 'Nov', 'transportation', 'performance', 'with', 'Rly', 'Board', 'and',
'Zonal', 'Rly', 'Officers', 'and', 'passedâ\x80']

```

```

SpaCy :  ['RT', '@RailMinIndia', ':', 'Member', 'Traffic', 'reviewed', 'the',

```

```
'April-', 'Nov', 'transportation', ' ', 'performance', 'with', 'Rly', 'Board',
'and', 'Zonal', 'Rly', 'Officers', 'and', 'passedâ\x80', '']
```

Tweet 4 :

```
NLTK : ['@', 'RailMinIndia', '@', 'sureshprabhu', 'TRAIN', 'NO', '12764',
'S13', 'No', 'water', 'please', 'address', 'finding', 'difficult', 'Seat', 'no',
'S13', '42,43,45,46,48']
```

```
SpaCy : ['@RailMinIndia', '@sureshprabhu', 'TRAIN', 'NO', '12764', 'S13',
'No', 'water', 'please', 'address', 'finding', 'difficult', 'Seat', 'no', 'S13',
'42,43,45,46,48']
```

Tweet 5 :

```
NLTK : ['@', 'RailMinIndia', 'my', 'pnr', 'no.2217356121', '.', 'pantry',
'charging', '20rs', 'for', 'water', 'bottle', 'which', 'has', 'printed', 'rate',
'of', '15']
```

```
SpaCy : ['@RailMinIndia', 'my', 'pnr', 'no.2217356121', '.', 'pantry',
'charging', '20rs', 'for', 'water', 'bottle', 'which', 'has', 'printed', 'rate',
'of', '15']
```

```
[52]: from nltk.stem import PorterStemmer

stemmer = PorterStemmer()

df['stemmed_text'] = df['clean_text'].apply(lambda x : ' '.join([stemmer.
    ↪stem(w) for w in x.split()])))
```

```
[58]: def lemmatize(text):
    doc = nlp(text)
    return ' '.join([token.lemma_ for token in doc])

df['lemmatized_text'] = df['clean_text'].apply(lemmatize)
```

```
[59]: def get_pos_tags(text):
    doc = nlp(text)
    return [(token.text, token.pos_, token.tag_) for token in doc]

df['pos_tags'] = df['clean_text'].apply(get_pos_tags)

df.head(20)
```

```
[59]:   index  sentiment_label      text \
1      1                0  @RailMinIndia My PNR is 8348062961, I am in wa...
2      2                0  @sureshprabhu @RailMinIndia AC not working in...
3      3                0  @RailMinIndia I'm traveling to chennai by trai...
4      4                5  @RailMinIndia irtc is not responding at the t...
5      5                7  @DRMbhopal @RailMinIndia @sanjaygupta2012 @drm...
6      6                6  @RailMinIndia If you can't give justice to gra...
7      7                4  @sureshprabhu @RailMinIndiaPlz wrkout smthng ...
```

8	8	0	@RailMinIndia @mumbairailusers Dirty water flo...
9	9	6	Enter to win \$150 Amazon Gift Card! #Books #PN...
10	10	3	@RailMinIndia still vendors are selling local ...
11	11	4	@Central_Railway @RailMinIndia @drmmumbaicr pl...
12	12	0	@RailMinIndia #Swatch Bharat Abhiyaan# Train s...
13	13	5	@RailMinIndia @sureshpprabhu my reserved ticke...
14	14	4	@RailMinIndia update:- today intercity 12930 ...
15	15	0	@RailMinIndia TRAIN NO 12656 THERE IS NO PROP...
16	16	6	The Google i1 button is used 5 billion times p...
17	17	3	@RailMinIndia Sir, ?? ?????? ?????? ?????? ?? ?...
18	18	4	@RailMinIndia Train has departed Varanasi and ...
19	19	4	@sureshpprabhu @RailMinIndia student's pnr 21...
20	20	5	@RailMinIndia Dear Sir, my father is a senior ...

	tweet_length		mentions \
1	140		[@RailMinIndia]
2	137		[@sureshpprabhu, @RailMinIndia]
3	122		[@RailMinIndia]
4	68		[@RailMinIndia]
5	104	[@DRMbhopal, @RailMinIndia, @sanjaygupta2012, ...	
6	127		[@RailMinIndia]
7	139		[@sureshpprabhu, @RailMinIndiaPlz]
8	131		[@RailMinIndia, @mumbairailusers]
9	94		[]
10	117		[@RailMinIndia]
11	120	[@Central_Railway, @RailMinIndia, @drmmumbaicr]	
12	140		[@RailMinIndia]
13	137		[@RailMinIndia, @sureshpprabhu]
14	58		[@RailMinIndia]
15	121		[@RailMinIndia]
16	74		[]
17	140		[@RailMinIndia]
18	103		[@RailMinIndia]
19	140		[@sureshpprabhu, @RailMinIndia]
20	140		[@RailMinIndia]

		hashtags	urls \
1		[]	[]
2		[]	[]
3		[]	[]
4		[]	[]
5		[]	[]
6		[]	[]
7		[]	[]
8		[]	[https://t.co/mBZUOFkLXQ]
9	[romance, books, urbanfantasy, pnr]		[https://t.co/jBfXzDe7P2]
10		[]	[https://t.co/wdFyLlmMek]

11		[]	[]
12	[no, swatch]	[https://t.co/h5uUdJuvel]	
13		[]	[]
14		[]	[]
15		[https://t.co/VIQmURi8HK]	
16		[]	[]
17		[https://t.co/sMfHUZdN2U]	
18		[]	[]
19		[]	[]
20		[]	[]

```

                                clean_text \
1  @RailMinIndia My PNR is 8348062961, I am in wa...
2  @sureshpprabhu @RailMinIndia AC not working in...
3  @RailMinIndia I'm traveling to chennai by trai...
4  @RailMinIndia irtc is not responding at the t...
5  @DRMBhopal @RailMinIndia @sanjaygupta2012 @drm...
6  @RailMinIndia If you can't give justice to gra...
7  @sureshpprabhu @RailMinIndiaPlz wrkout smthng ...
8  @RailMinIndia @mumbairailusers Dirty water flo...
9  Enter to win $150 Amazon Gift Card! #Books #PN...
10 @RailMinIndia still vendors are selling local ...
11 @Central_Railway @RailMinIndia @drmmumbaicr pl...
12 @RailMinIndia #Swatch Bharat Abhiyaan# Train s...
13 @RailMinIndia @sureshpprabhu my reserved ticke...
14 @RailMinIndia update:- today intercity 12930 ...
15 @RailMinIndia TRAIN NO 12656 THERE IS NO PROP...
16 The Google il button is used 5 billion times p...
17 @RailMinIndia Sir, ?? ?????? ?????? ?????? ?? ?...
18 @RailMinIndia Train has departed Varanasi and ...
19 @sureshpprabhu @RailMinIndia student's pnr 21...
20 @RailMinIndia Dear Sir, my father is a senior ...

```

```

                                stemmed_text \
1  @railminindia my pnr is 8348062961, i am in wa...
2  @sureshpprabhu @railminindia ac not work in pr...
3  @railminindia i'm travel to chennai by train 1...
4  @railminindia irtc is not respond at the time...
5  @drmbhopal @railminindia @sanjaygupta2012 @drm...
6  @railminindia if you can't give justic to grad...
7  @sureshpprabhu @railminindiaplz wrkout smthng ...
8  @railminindia @mumbairailus dirti water flow f...
9  enter to win $150 amazon gift card! #book #pnr...
10 @railminindia still vendor are sell local loca...
11 @central_railway @railminindia @drmmumbaicr pl...
12 @railminindia #swatch bharat abhiyaan# train s...
13 @railminindia @sureshpprabhu my reserv ticket ...

```

14 @railminindia update:- today interc 12930 late...
 15 @railminindia train no 12656 there is no prope...
 16 the googl i1 button is use 5 billion time pnr ...
 17 @railminindia sir, ?? ?????? ?????? ?????? ?? ?...
 18 @railminindia train ha depart varanasi and wil...
 19 @sureshpprabhu @railminindia student' pnr 2119...
 20 @railminindia dear sir, my father is a senior ...

lemmatized_text \

1 @RailMinIndia my PNR be 8348062961 , I be in w...
 2 @sureshpprabhu @railminindia AC not work in pr...
 3 @railminindia I be travel to chennai by train ...
 4 @RailMinIndia irttc be not respond at the time...
 5 @DRMbhopal @railminindia @sanjaygupta2012 @drm...
 6 @RailMinIndia if you can not give justice to g...
 7 @sureshpprabhu @RailMinIndiaPlz wrkout smthng ...
 8 @RailMinIndia @mumbairailusers Dirty water flo...
 9 enter to win \$ 150 Amazon Gift Card ! # Books ...
 10 @RailMinIndia still vendor be sell local local...
 11 @central_railway @RailMinIndia @drmmumbaicr pl...
 12 @RailMinIndia # Swatch Bharat Abhiyaan # Train...
 13 @RailMinIndia @sureshpprabhu my reserved ticke...
 14 @RailMinIndia update:- today intercity 12930...
 15 @RailMinIndia TRAIN no 12656 there be no pro...
 16 the Google i1 button be use 5 billion time pnr...
 17 @RailMinIndia Sir , ? ? ? ? ? ? ? ? ? ? ? ? ? ...
 18 @RailMinIndia Train have depart Varanasi and w...
 19 @sureshpprabhu @RailMinIndia student 's pnr ...
 20 @RailMinIndia Dear Sir , my father be a senior...

pos_tags

1 [(@RailMinIndia, NOUN, NNS), (My, PRON, PRP\$),...
 2 [(@sureshpprabhu, NOUN, NN), (@RailMinIndia, N...
 3 [(@RailMinIndia, NOUN, NN), (I, PRON, PRP), ('...
 4 [(@RailMinIndia, PROPN, NNP), (irttc, NOUN, NN...
 5 [(@DRMbhopal, PROPN, NNP), (@RailMinIndia, NOU...
 6 [(@RailMinIndia, NOUN, NNS), (If, CONJ, IN), ...
 7 [(@sureshpprabhu, PROPN, NNP), (@RailMinIndiaP...
 8 [(@RailMinIndia, PROPN, NNP), (@mumbairailuser...
 9 [(Enter, VERB, VB), (to, PART, TO), (win, VERB...
 10 [(@RailMinIndia, PROPN, NNP), (still, ADV, RB)...
 11 [(@Central_Railway, NOUN, NN), (@RailMinIndia,...
 12 [(@RailMinIndia, NOUN, NNS), (#, SYM, \$), (Swa...
 13 [(@RailMinIndia, NOUN, NNS), (@sureshpprabhu, ...
 14 [(@RailMinIndia, PROPN, NNP), (, SPACE, _SP),...
 15 [(@RailMinIndia, NOUN, NNS), (, SPACE, _SP), ...
 16 [(The, DET, DT), (Google, PROPN, NNP), (i1, PR...

```

17 [(@RailMinIndia, PROPN, NNP), (Sir, PROPN, NNP...
18 [(@RailMinIndia, PROPN, NNP), (Train, PROPN, N...
19 [(@sureshpprabhu, NOUN, NN), ( , SPACE, _SP), ...
20 [(@RailMinIndia, NOUN, NNS), (Dear, PROPN, NNP...

```

```
[60]: df.iloc[0]
```

```

[60]: index                                1
      sentiment_label                      0
      text                @RailMinIndia My PNR is 8348062961, I am in wa...
      tweet_length                                140
      mentions                                [@RailMinIndia]
      hastags                                []
      urls                                    []
      clean_text                @RailMinIndia My PNR is 8348062961, I am in wa...
      stemmed_text              @railminindia my pnr is 8348062961, i am in wa...
      lemmatized_text           @RailMinIndia my PNR be 8348062961 , I be in w...
      pos_tags                  [(@RailMinIndia, NOUN, NNS), (My, PRON, PRP$),...
      Name: 1, dtype: object

```

```

[62]: import time

start_time = time.time()

for _ in range(100):
    lemmatize(df.iloc[0]['clean_text'])
end_time = time.time()

avg_time_per_tweet = (end_time - start_time) / 100
print(f"Average processing time : {avg_time_per_tweet : .4f} seconds")

```

Average processing time : 0.0067 seconds

7 6 NamedEntity Recognition

```

[64]: def extract_entites(text):
      doc = nlp(text)
      return [(ent.text, ent.label_) for ent in doc.ents]

df['entites'] = df['clean_text'].apply(extract_entites)
df.head()

```

```

[64]:   index  sentiment_label                                text \
1      1                  0 @RailMinIndia My PNR is 8348062961, I am in wa...
2      2                  0 @sureshpprabhu @RailMinIndia AC not working in...
3      3                  0 @RailMinIndia I'm traveling to chennai by trai...
4      4                  5 @RailMinIndia irtc is not responding at the t...

```

```

5      5      7 @DRMbhopal @RailMinIndia @sanjaygupta2012 @drm...

tweet_length      mentions hastags \
1      140      [@RailMinIndia]      []
2      137      [@sureshpprabhu, @RailMinIndia]      []
3      122      [@RailMinIndia]      []
4      68      [@RailMinIndia]      []
5      104      [@DRMbhopal, @RailMinIndia, @sanjaygupta2012, ...      []

urls      clean_text \
1      []      @RailMinIndia My PNR is 8348062961, I am in wa...
2      []      @sureshpprabhu @RailMinIndia AC not working in...
3      []      @RailMinIndia I'm traveling to chennai by trai...
4      []      @RailMinIndia irttc is not responding at the t...
5      []      @DRMbhopal @RailMinIndia @sanjaygupta2012 @drm...

stemmed_text \
1 @railminindia my pnr is 8348062961, i am in wa...
2 @sureshpprabhu @railminindia ac not work in pr...
3 @railminindia i'm travel to chennai by train 1...
4 @railminindia irttc is not respond at the time...
5 @drmbhopal @railminindia @sanjaygupta2012 @drm...

lemmatized_text \
1 @RailMinIndia my PNR be 8348062961 , I be in w...
2 @sureshpprabhu @railminindia AC not work in pr...
3 @railminindia I be travel to chennai by train ...
4 @RailMinIndia irttc be not respond at the time...
5 @DRMbhopal @railminindia @sanjaygupta2012 @drm...

pos_tags \
1 [(@RailMinIndia, NOUN, NNS), (My, PRON, PRP$),...]
2 [(@sureshpprabhu, NOUN, NN), (@RailMinIndia, N...
3 [(@RailMinIndia, NOUN, NN), (I, PRON, PRP), ('...
4 [(@RailMinIndia, PROPN, NNP), (irttc, NOUN, NN...
5 [(@DRMbhopal, PROPN, NNP), (@RailMinIndia, NOU...

entites
1      [(PNR, ORG), (8348062961, DATE)]
2      [(@sureshpprabhu @RailMinIndia AC, ORG), (toda...
3      [(16102, CARDINAL), (S4-46, DATE), (one, CARDI...
4      []
5      []

```

```

[77]: def map_domain_entites(entites):
      mapped = []
      for text, label in entites:

```

```

    if label == 'GPE':
        mapped.append((text, 'STATION'))
    elif label == 'CARDINAL' and len(text) > 2:
        mapped.append((text, 'TRAIN_NO'))
    elif label == 'ORDINAL':
        mapped.append((text, 'COACH'))
    elif text.lower in ['water', 'ac', 'cleanliness', 'toilet']:
        mapped.append((text, 'SERVICE'))
    else:
        mapped.append((text, 'OTHER'))

df['domain_entites'] = df['entites'].apply(map_domain_entites)

df.head(20)

```

```

[77]:      index sentiment_label      text \
1         1              0  @RailMinIndia My PNR is 8348062961, I am in wa...
2         2              0  @sureshprabhu @RailMinIndia AC not working in...
3         3              0  @RailMinIndia I'm traveling to chennai by trai...
4         4              5  @RailMinIndia irtc is not responding at the t...
5         5              7  @DRMbhopal @RailMinIndia @sanjaygupta2012 @drmm...
6         6              6  @RailMinIndia If you can't give justice to gra...
7         7              4  @sureshprabhu @RailMinIndiaPlz wrkout smthng ...
8         8              0  @RailMinIndia @mumbairailusers Dirty water flo...
9         9              6  Enter to win $150 Amazon Gift Card! #Books #PN...
10        10              3  @RailMinIndia still vendors are selling local ...
11        11              4  @Central_Railway @RailMinIndia @drmmumbaicr pl...
12        12              0  @RailMinIndia #Swatch Bharat Abhiyaan# Train s...
13        13              5  @RailMinIndia @sureshprabhu my reserved ticke...
14        14              4  @RailMinIndia  update:- today intercity 12930 ...
15        15              0  @RailMinIndia  TRAIN NO 12656 THERE IS NO PROP...
16        16              6  The Google il button is used 5 billion times p...
17        17              3  @RailMinIndia Sir, ?? ?????? ?????? ?????? ?? ?...
18        18              4  @RailMinIndia Train has departed Varanasi and ...
19        19              4  @sureshprabhu  @RailMinIndia student's pnr 21...
20        20              5  @RailMinIndia Dear Sir, my father is a senior ...

```

```

      tweet_length      mentions \
1          140      [@RailMinIndia]
2          137      [@sureshprabhu, @RailMinIndia]
3          122      [@RailMinIndia]
4           68      [@RailMinIndia]
5          104  [@DRMbhopal, @RailMinIndia, @sanjaygupta2012, ...
6          127      [@RailMinIndia]
7          139      [@sureshprabhu, @RailMinIndiaPlz]
8          131      [@RailMinIndia, @mumbairailusers]
9           94      []

```


10	117	[@RailMinIndia]
11	120	[@Central_Railway, @RailMinIndia, @drmmumbaicr]
12	140	[@RailMinIndia]
13	137	[@RailMinIndia, @sureshpprabhu]
14	58	[@RailMinIndia]
15	121	[@RailMinIndia]
16	74	[]
17	140	[@RailMinIndia]
18	103	[@RailMinIndia]
19	140	[@sureshpprabhu, @RailMinIndia]
20	140	[@RailMinIndia]

	hashtags	urls \
1	[]	[]
2	[]	[]
3	[]	[]
4	[]	[]
5	[]	[]
6	[]	[]
7	[]	[]
8	[]	[https://t.co/mBZUOfkLXQ]
9	[romance, books, urbanfantasy, pnr]	[https://t.co/jBfXzDe7P2]
10	[]	[https://t.co/wdFyLlmMek]
11	[]	[]
12	[no, swatch]	[https://t.co/h5uUdJuvel]
13	[]	[]
14	[]	[]
15	[]	[https://t.co/VIQmURi8HK]
16	[]	[]
17	[]	[https://t.co/sMfHUZdN2U]
18	[]	[]
19	[]	[]
20	[]	[]

	clean_text \
1	@RailMinIndia My PNR is 8348062961, I am in wa...
2	@sureshpprabhu @RailMinIndia AC not working in...
3	@RailMinIndia I'm traveling to chennai by trai...
4	@RailMinIndia irctc is not responding at the t...
5	@DRMbhopal @RailMinIndia @sanjaygupta2012 @drm...
6	@RailMinIndia If you can't give justice to gra...
7	@sureshpprabhu @RailMinIndiaPlz wrkout smthng ...
8	@RailMinIndia @mumbairailusers Dirty water flo...
9	Enter to win \$150 Amazon Gift Card! #Books #PN...
10	@RailMinIndia still vendors are selling local ...
11	@Central_Railway @RailMinIndia @drmmumbaicr pl...
12	@RailMinIndia #Swatch Bharat Abhiyaan# Train s...

13 @RailMinIndia @sureshprabhu my reserved ticke...
 14 @RailMinIndia update:- today intercity 12930 ...
 15 @RailMinIndia TRAIN NO 12656 THERE IS NO PROP...
 16 The Google i1 button is used 5 billion times p...
 17 @RailMinIndia Sir, ?? ?????? ?????? ?????? ?? ?...
 18 @RailMinIndia Train has departed Varanasi and ...
 19 @sureshprabhu @RailMinIndia student's pnr 21...
 20 @RailMinIndia Dear Sir, my father is a senior ...

stemmed_text \

1 @railminindia my pnr is 8348062961, i am in wa...
 2 @sureshprabhu @railminindia ac not work in pr...
 3 @railminindia i'm travel to chennai by train 1...
 4 @railminindia irttc is not respond at the time...
 5 @drmbhopal @railminindia @sanjaygupta2012 @drm...
 6 @railminindia if you can't give justic to grad...
 7 @sureshprabhu @railminindiaplz wrkout smthng ...
 8 @railminindia @mumbairailus dirti water flow f...
 9 enter to win \$150 amazon gift card! #book #pnr...
 10 @railminindia still vendor are sell local loca...
 11 @central_railway @railminindia @drmmumbaicr pl...
 12 @railminindia #swatch bharat abhiyaan# train s...
 13 @railminindia @sureshprabhu my reserv ticket ...
 14 @railminindia update:- today interc 12930 late...
 15 @railminindia train no 12656 there is no prope...
 16 the googl i1 button is use 5 billion time pnr ...
 17 @railminindia sir, ?? ?????? ?????? ?????? ?? ?...
 18 @railminindia train ha depart varanasi and wil...
 19 @sureshprabhu @railminindia student' pnr 2119...
 20 @railminindia dear sir, my father is a senior ...

lemmatized_text \

1 @RailMinIndia my PNR be 8348062961 , I be in w...
 2 @sureshprabhu @railminindia AC not work in pr...
 3 @railminindia I be travel to chennai by train ...
 4 @RailMinIndia irttc be not respond at the time...
 5 @DRMbhopal @railminindia @sanjaygupta2012 @drm...
 6 @RailMinIndia if you can not give justice to g...
 7 @sureshprabhu @RailMinIndiaPlz wrkout smthng ...
 8 @RailMinIndia @mumbairailusers Dirty water flo...
 9 enter to win \$ 150 Amazon Gift Card ! # Books ...
 10 @RailMinIndia still vendor be sell local local...
 11 @central_railway @RailMinIndia @drmmumbaicr pl...
 12 @RailMinIndia # Swatch Bharat Abhiyaan # Train...
 13 @RailMinIndia @sureshprabhu my reserved ticke...
 14 @RailMinIndia update:- today intercity 12930...
 15 @RailMinIndia TRAIN no 12656 there be no pro...

16 the Google i1 button be use 5 billion time pnr...
 17 @RailMinIndia Sir , ? ? ? ? ? ? ? ? ? ? ? ? ? ...
 18 @RailMinIndia Train have depart Varanasi and w...
 19 @sureshpprabhu @RailMinIndia student 's pnr ...
 20 @RailMinIndia Dear Sir , my father be a senior...

pos_tags \

1 [(@RailMinIndia, NOUN, NNS), (My, PRON, PRP\$),...
 2 [(@sureshpprabhu, NOUN, NN), (@RailMinIndia, N...
 3 [(@RailMinIndia, NOUN, NN), (I, PRON, PRP), ('...
 4 [(@RailMinIndia, PROPN, NNP), (irctc, NOUN, NN...
 5 [(@DRMbhupal, PROPN, NNP), (@RailMinIndia, NOU...
 6 [(@RailMinIndia, NOUN, NNS), (If, CONJ, IN), ...
 7 [(@sureshpprabhu, PROPN, NNP), (@RailMinIndiaP...
 8 [(@RailMinIndia, PROPN, NNP), (@mumbairailuser...
 9 [(Enter, VERB, VB), (to, PART, TO), (win, VERB...
 10 [(@RailMinIndia, PROPN, NNP), (still, ADV, RB)...
 11 [(@Central_Railway, NOUN, NN), (@RailMinIndia,...
 12 [(@RailMinIndia, NOUN, NNS), (#, SYM, \$), (Swa...
 13 [(@RailMinIndia, NOUN, NNS), (@sureshpprabhu, ...
 14 [(@RailMinIndia, PROPN, NNP), (, SPACE, _SP),...
 15 [(@RailMinIndia, NOUN, NNS), (, SPACE, _SP), ...
 16 [(The, DET, DT), (Google, PROPN, NNP), (i1, PR...
 17 [(@RailMinIndia, PROPN, NNP), (Sir, PROPN, NNP...
 18 [(@RailMinIndia, PROPN, NNP), (Train, PROPN, N...
 19 [(@sureshpprabhu, NOUN, NN), (, SPACE, _SP), ...
 20 [(@RailMinIndia, NOUN, NNS), (Dear, PROPN, NNP...

	entites	domain_entites
1	[(PNR, ORG), (8348062961, DATE)]	None
2	[(@sureshpprabhu @RailMinIndia AC, ORG), (toda...	None
3	[(16102, CARDINAL), (S4-46, DATE), (one, CARDI...	None
4	[]	None
5	[]	None
6	[(one, CARDINAL), (SSE, ORG)]	None
7	[(@sureshpprabhu @RailMinIndiaPlz wrkout, ORG)...	None
8	[(8.08, CARDINAL)]	None
9	[(150, MONEY), (Amazon, ORG), (Gift Card, PERS...	None
10	[]	None
11	[(@Central_Railway @RailMinIndia, ORG), (12141...	None
12	[(Swatch Bharat Abhiyaan, ORG), (12218, DATE),...	None
13	[(RAC, ORG)]	None
14	[(today, DATE), (12930, DATE), (16, CARDINAL)]	None
15	[]	None
16	[(Google, ORG), (5 billion, CARDINAL), (AllTwi...	None
17	[]	None
18	[(@RailMinIndia Train, ORG), (Varanasi, LOC), ...	None

```

19 [(@sureshpprabhu @RailMinIndia, ORG), (211902... None
20 [(RAC, ORG)] None

```

```

[84]: import pandas as pd
from vaderSentiment.vaderSentiment import SentimentIntensityAnalyzer, negated
from afinn import Afinn
from nrclex import NRCLex

#initialing sentiment tools
vader_analyzer = SentimentIntensityAnalyzer()
afinn = Afinn()

def clean_text(text):
    return text.replace("RT", "").replace(r"@S+", "").replace(r"#S+", "").strip()

df['clean_text2'] = df['text'].apply(clean_text)

#compute sentiment scores

df['vader_score'] = df['clean_text2'].apply(lambda x : vader_analyzer.
    ↪polarity_scores(x)['compound'])
df['affin_score'] = df['clean_text2'].apply(lambda x : afinn.score(x))

def get_nrc_score(text):
    try:
        emotion = NRCLex(text = text)
        return sum(emotion.affect_frequencies.values())
    except Exception as e:
        return 0

df['nrc_score'] = df['clean_text'].apply(get_nrc_score)

threshold = 0.05

df['vader_polarity'] = df['vader_score'].apply(lambda x : 'positive' if x >= threshold
    ↪else 'negative')
df['affin_polarity'] = df['affin_score'].apply(lambda x : 'positive' if x > 0
    ↪else 'negative')
df['nrc_polarity'] = df['nrc_score'].apply(lambda x : 'positive' if x > 0 else
    ↪'negative')

#compute alignment
df['alignment'] = df.apply(
    lambda row : 1 if row['vader_polarity'] == row['affin_polarity'] ==
    ↪row['nrc_polarity'] else 0,

```

```

        axis=1
    )

    #show some aligned vs non-aligned rows
    aligned_df = df[df['alignment'] == 1]
    non_aligned_df = df[df['alignment'] == 0]

    # print result
    total = len(df)
    aligned = len(aligned_df)
    agreement_rate = aligned / total*100

    print(f"\nTotal Tweets : {total}")
    print(f"Aligned Tweets : {aligned}")
    print(f"Agreement rate : {agreement_rate : .2f}%")

```

```

Total Tweets : 1366
Aligned Tweets : 432
Agreement rate : 31.63%

```

```
[85]: df.head()
```

```

[85]:   index sentiment_label      text \
1      1              0  @RailMinIndia My PNR is 8348062961, I am in wa...
2      2              0  @sureshpprabhu @RailMinIndia AC not working in...
3      3              0  @RailMinIndia I'm traveling to chennai by trai...
4      4              5  @RailMinIndia irtc is not responding at the t...
5      5              7  @DRMbhopal @RailMinIndia @sanjaygupta2012 @drm...

      tweet_length      mentions hastags \
1           140      [@RailMinIndia]    []
2           137  [@sureshpprabhu, @RailMinIndia]  []
3           122      [@RailMinIndia]    []
4            68      [@RailMinIndia]    []
5           104  [@DRMbhopal, @RailMinIndia, @sanjaygupta2012, ...]  []

      urls      clean_text \
1  []  @RailMinIndia My PNR is 8348062961, I am in wa...
2  []  @sureshpprabhu @RailMinIndia AC not working in...
3  []  @RailMinIndia I'm traveling to chennai by trai...
4  []  @RailMinIndia irtc is not responding at the t...
5  []  @DRMbhopal @RailMinIndia @sanjaygupta2012 @drm...

      stemmed_text \
1  @railminindia my pnr is 8348062961, i am in wa...

```

```

2 @sureshpprabhu @railminindia ac not work in pr...
3 @railminindia i'm travel to chennai by train 1...
4 @railminindia irctc is not respond at the time...
5 @drmbhopal @railminindia @sanjaygupta2012 @drm...

```

```

                                lemmatized_text  ...  \
1  @RailMinIndia my PNR be 8348062961 , I be in w...  ...
2  @sureshpprabhu @railminindia AC not work in pr...  ...
3  @railminindia I be travel to chennai by train ...  ...
4  @RailMinIndia irctc be not respond at the time...  ...
5  @DRMbhopal @railminindia @sanjaygupta2012 @drm...  ...

```

```

                                entites domain_entites  \
1                                [(PNR, ORG), (8348062961, DATE)]      None
2  [(@sureshpprabhu @RailMinIndia AC, ORG), (toda...      None
3  [(16102, CARDINAL), (S4-46, DATE), (one, CARDI...      None
4                                []                                     None
5                                []                                     None

```

```

                                clean_text2 vader_score  affin_score  \
1  @RailMinIndia My PNR is 8348062961, I am in wa...      -0.4767      -2.0
2  @sureshpprabhu @RailMinIndia AC not working in...      -0.2235      -1.0
3  @RailMinIndia I'm traveling to chennai by trai...       0.0000      -2.0
4  @RailMinIndia irctc is not responding at the t...       0.0000       0.0
5  @DRMbhopal @RailMinIndia @sanjaygupta2012 @drm...       0.0258      -1.0

```

```

nrc_score  vader_polarity affin_polarity nrc_polarity alignment
1          1.0          negative          negative          positive          0
2          1.0          negative          negative          positive          0
3          1.0          negative          negative          positive          0
4          1.0          negative          negative          positive          0
5          1.0          negative          negative          positive          0

```

[5 rows x 21 columns]

```

[96]: #dependency chunk - liked/disliked clause extractor
import spacy
nlp = spacy.load('en_core_web_sm')

def extract_clause(text, postive=True):
    doc = nlp(text)
    for sent in doc.sents:
        if postive and any(tok.lemma_ in {"good", "great", "thank"} for tok in
↪sent):
            return sent.text.strip()
        if not postive and any(tok.lemma_ in {"bad", "not", "delay", "poor",
↪"dirty"} for tok in sent):

```

```
return sent.text.strip()
```

```
[97]: #filter 10 random positive and 10 negative tweets
pos_tweets = df[df['vader_polarity'] == "positive"].sample(10, random_state=42)
neg_tweets = df[df['vader_polarity'] == "negative"].sample(10, random_state=42)

#extract liked/disliked clause

pos_tweets["liked_clause"] = pos_tweets['clean_text'].apply(lambda x :
    ↪extract_clause(x, True))
neg_tweets['disliked_clause'] = neg_tweets['clean_text'].apply(lambda x :
    ↪extract_clause(x, False))
```

```
[98]: df.head(20)
```

```
[98]:
```

	index	sentiment_label	text \
1	1	0	@RailMinIndia My PNR is 8348062961, I am in wa...
2	2	0	@sureshprabhu @RailMinIndia AC not working in...
3	3	0	@RailMinIndia I'm traveling to chennai by trai...
4	4	5	@RailMinIndia irtc is not responding at the t...
5	5	7	@DRMbhopal @RailMinIndia @sanjaygupta2012 @drm...
6	6	6	@RailMinIndia If you can't give justice to gra...
7	7	4	@sureshprabhu @RailMinIndiaPlz wrkout smthng ...
8	8	0	@RailMinIndia @mumbairailusers Dirty water flo...
9	9	6	Enter to win \$150 Amazon Gift Card! #Books #PN...
10	10	3	@RailMinIndia still vendors are selling local ...
11	11	4	@Central_Railway @RailMinIndia @drmmumbaicr pl...
12	12	0	@RailMinIndia #Swatch Bharat Abhiyaan# Train s...
13	13	5	@RailMinIndia @sureshprabhu my reserved ticke...
14	14	4	@RailMinIndia update:- today intercity 12930 ...
15	15	0	@RailMinIndia TRAIN NO 12656 THERE IS NO PROP...
16	16	6	The Google il button is used 5 billion times p...
17	17	3	@RailMinIndia Sir, ?? ?????? ?????? ?????? ?? ?...
18	18	4	@RailMinIndia Train has departed Varanasi and ...
19	19	4	@sureshprabhu @RailMinIndia student's pnr 21...
20	20	5	@RailMinIndia Dear Sir, my father is a senior ...

	tweet_length	mentions \
1	140	[@RailMinIndia]
2	137	[@sureshprabhu, @RailMinIndia]
3	122	[@RailMinIndia]
4	68	[@RailMinIndia]
5	104	[@DRMbhopal, @RailMinIndia, @sanjaygupta2012, ...
6	127	[@RailMinIndia]
7	139	[@sureshprabhu, @RailMinIndiaPlz]
8	131	[@RailMinIndia, @mumbairailusers]
9	94	[]

10	117		[@RailMinIndia]
11	120	[@Central_Railway, @RailMinIndia, @drmmumbaicr]	
12	140		[@RailMinIndia]
13	137	[@RailMinIndia, @sureshpprabhu]	
14	58		[@RailMinIndia]
15	121		[@RailMinIndia]
16	74		[]
17	140		[@RailMinIndia]
18	103		[@RailMinIndia]
19	140	[@sureshpprabhu, @RailMinIndia]	
20	140		[@RailMinIndia]

	hashtags	urls \
1	[]	[]
2	[]	[]
3	[]	[]
4	[]	[]
5	[]	[]
6	[]	[]
7	[]	[]
8	[]	[https://t.co/mBZUOfkLXQ]
9	[romance, books, urbanfantasy, pnr]	[https://t.co/jBfXzDe7P2]
10	[]	[https://t.co/wdFyLlmMek]
11	[]	[]
12	[no, swatch]	[https://t.co/h5uUdJuvel]
13	[]	[]
14	[]	[]
15	[]	[https://t.co/VIQmURi8HK]
16	[]	[]
17	[]	[https://t.co/sMfHUZdN2U]
18	[]	[]
19	[]	[]
20	[]	[]

	clean_text \
1	@RailMinIndia My PNR is 8348062961, I am in wa...
2	@sureshpprabhu @RailMinIndia AC not working in...
3	@RailMinIndia I'm traveling to chennai by trai...
4	@RailMinIndia irttc is not responding at the t...
5	@DRMbhopal @RailMinIndia @sanjaygupta2012 @drm...
6	@RailMinIndia If you can't give justice to gra...
7	@sureshpprabhu @RailMinIndiaPlz wrkout smthng ...
8	@RailMinIndia @mumbairailusers Dirty water flo...
9	Enter to win \$150 Amazon Gift Card! #Books #PN...
10	@RailMinIndia still vendors are selling local ...
11	@Central_Railway @RailMinIndia @drmmumbaicr pl...
12	@RailMinIndia #Swatch Bharat Abhiyaan# Train s...

13 @RailMinIndia @sureshprabhu my reserved ticke...
 14 @RailMinIndia update:- today intercity 12930 ...
 15 @RailMinIndia TRAIN NO 12656 THERE IS NO PROP...
 16 The Google i1 button is used 5 billion times p...
 17 @RailMinIndia Sir, ?? ?????? ?????? ?????? ?? ?...
 18 @RailMinIndia Train has departed Varanasi and ...
 19 @sureshprabhu @RailMinIndia student's pnr 21...
 20 @RailMinIndia Dear Sir, my father is a senior ...

stemmed_text \

1 @railminindia my pnr is 8348062961, i am in wa...
 2 @sureshprabhu @railminindia ac not work in pr...
 3 @railminindia i'm travel to chennai by train 1...
 4 @railminindia irttc is not respond at the time...
 5 @drmbhopal @railminindia @sanjaygupta2012 @drm...
 6 @railminindia if you can't give justic to grad...
 7 @sureshprabhu @railminindiaplz wrkout smthng ...
 8 @railminindia @mumbairailus dirti water flow f...
 9 enter to win \$150 amazon gift card! #book #pnr...
 10 @railminindia still vendor are sell local loca...
 11 @central_railway @railminindia @drmmumbaicr pl...
 12 @railminindia #swatch bharat abhiyaan# train s...
 13 @railminindia @sureshprabhu my reserv ticket ...
 14 @railminindia update:- today interc 12930 late...
 15 @railminindia train no 12656 there is no prope...
 16 the googl i1 button is use 5 billion time pnr ...
 17 @railminindia sir, ?? ?????? ?????? ?????? ?? ?...
 18 @railminindia train ha depart varanasi and wil...
 19 @sureshprabhu @railminindia student' pnr 2119...
 20 @railminindia dear sir, my father is a senior ...

lemmatized_text ... \

1 @RailMinIndia my PNR be 8348062961 , I be in w... ...
 2 @sureshprabhu @railminindia AC not work in pr... ...
 3 @railminindia I be travel to chennai by train
 4 @RailMinIndia irttc be not respond at the time... ...
 5 @DRMbhopal @railminindia @sanjaygupta2012 @drm... ...
 6 @RailMinIndia if you can not give justice to g... ...
 7 @sureshprabhu @RailMinIndiaPlz wrkout smthng
 8 @RailMinIndia @mumbairailusers Dirty water flo... ...
 9 enter to win \$ 150 Amazon Gift Card ! # Books
 10 @RailMinIndia still vendor be sell local local... ...
 11 @central_railway @RailMinIndia @drmmumbaicr pl... ...
 12 @RailMinIndia # Swatch Bharat Abhiyaan # Train... ...
 13 @RailMinIndia @sureshprabhu my reserved ticke... ...
 14 @RailMinIndia update:- today intercity 12930... ...
 15 @RailMinIndia TRAIN no 12656 there be no pro... ...

16 the Google i1 button be use 5 billion time pnr... ...
 17 @RailMinIndia Sir , ? ? ? ? ? ? ? ? ? ? ? ? ?
 18 @RailMinIndia Train have depart Varanasi and w... ...
 19 @sureshp Prabhu @RailMinIndia student 's pnr
 20 @RailMinIndia Dear Sir , my father be a senior... ...

	clean_text2	vader_score	affin_score	\
1	@RailMinIndia My PNR is 8348062961, I am in wa...	-0.4767	-2.0	
2	@sureshp Prabhu @RailMinIndia AC not working in...	-0.2235	-1.0	
3	@RailMinIndia I'm traveling to chennai by trai...	0.0000	-2.0	
4	@RailMinIndia irtc is not responding at the t...	0.0000	0.0	
5	@DRMBhopal @RailMinIndia @sanjaygupta2012 @drm...	0.0258	-1.0	
6	@RailMinIndia If you can't give justice to gra...	-0.6093	1.0	
7	@sureshp Prabhu @RailMinIndiaPlz wrkout smthng ...	-0.2263	-1.0	
8	@RailMinIndia @mumbairailusers Dirty water flo...	-0.4404	-2.0	
9	Enter to win \$150 Amazon Gift Card! #Books #PN...	0.9060	8.0	
10	@RailMinIndia still vendors are selling local ...	0.3612	2.0	
11	@Central_Railway @RailMinIndia @drmmumbaicr pl...	0.3182	1.0	
12	@RailMinIndia #Swatch Bharat Abhiyaan# Train s...	-0.4767	-3.0	
13	@RailMinIndia @sureshp Prabhu my reserved ticke...	0.8205	3.0	
14	@RailMinIndia update:- today intercity 12930 ...	0.0000	0.0	
15	@RailMinIndia TRAIN NO 12656 THERE IS NO PROP...	-0.0258	-1.0	
16	The Google i1 button is used 5 billion times p...	0.0000	0.0	
17	@RailMinIndia Sir, ?? ?????? ?????? ?????? ?? ?...	0.0000	0.0	
18	@RailMinIndia Train has departed Varanasi and ...	0.2023	1.0	
19	@sureshp Prabhu @RailMinIndia student's pnr 21...	-0.3182	-1.0	
20	@RailMinIndia Dear Sir, my father is a senior ...	0.0772	0.0	

	nrc_score	vader_polarity	affin_polarity	nrc_polarity	alignment	\
1	1.0	negative	negative	positive	0	
2	1.0	negative	negative	positive	0	
3	1.0	negative	negative	positive	0	
4	1.0	negative	negative	positive	0	
5	1.0	negative	negative	positive	0	
6	1.0	negative	positive	positive	0	
7	1.0	negative	negative	positive	0	
8	1.0	negative	negative	positive	0	
9	0.0	positive	positive	negative	0	
10	0.0	positive	positive	negative	0	
11	1.0	positive	positive	positive	1	
12	0.0	negative	negative	negative	1	
13	1.0	positive	positive	positive	1	
14	1.0	negative	negative	positive	0	
15	0.0	negative	negative	negative	1	
16	0.0	negative	negative	negative	1	
17	0.0	negative	negative	negative	1	
18	1.0	positive	positive	positive	1	

19	1.0	negative	negative	positive	0
20	1.0	positive	negative	positive	0

	liked_clause	disliked_clause
1	NaN	NaN
2	NaN	NaN
3	NaN	NaN
4	NaN	NaN
5	NaN	NaN
6	NaN	NaN
7	NaN	NaN
8	NaN	NaN
9	NaN	NaN
10	NaN	NaN
11	NaN	NaN
12	NaN	NaN
13	NaN	NaN
14	NaN	NaN
15	NaN	NaN
16	NaN	NaN
17	NaN	NaN
18	NaN	NaN
19	NaN	NaN
20	NaN	NaN

[20 rows x 23 columns]

8 image processing

```
[103]: import os
import numpy as np
import matplotlib.pyplot as plt
from skimage import io, transform, filters, color, exposure
from pathlib import Path

def preprocess_images_skimage(data_dir, output_dir):
    """
    Preprocesses images in the specified data directory using scikit-image.
    Steps include: grayscale conversion, resizing, and histogram equalization.
    Processed images are saved to the output directory.

    Args:
        data_dir (str): Path to the input directory containing subfolders (e.g.
        ↪, '0', '1', ...)
                               which in turn contain emotion images (e.g., 'Happy.
        ↪.jpg').
```

```

    output_dir (str): Path to the directory where processed images will be
    ↪ saved.
    """
    # Create output directory if it does not exist
    Path(output_dir).mkdir(parents=True, exist_ok=True)

    emotions = ['Anger', 'Contempt', 'Disgust', 'Fear', 'Happy', 'Sad',
    ↪ 'Surprised', 'Neutral']

    # Define target image size for resizing (e.g., 100x100 pixels)
    img_size = (100, 100)

    # Iterate through potential subject folders (numbered 0 to 18)
    for folder_num in range(19): # Assuming subject folders are named 0, 1, ...
    ↪, 18
        folder_path = os.path.join(data_dir, str(folder_num))

        # Check if the current path is actually a directory
        if not os.path.isdir(folder_path):
            # print(f"Skipping {folder_path} as it is not a directory.") #
    ↪ Optional: uncomment for verbose skipping
            continue

        # Create corresponding output subdirectory for the current subject
    ↪ folder
        output_folder_path = Path(output_dir) / str(folder_num)
        output_folder_path.mkdir(parents=True, exist_ok=True)

        # Process each emotion's image within the current subject folder
        for emotion in emotions:
            img_filename = f"{emotion}.jpg"
            img_path = os.path.join(folder_path, img_filename)

            # Check if the image file exists
            if not os.path.exists(img_path):
                # print(f"Image not found: {img_path}. Skipping.") # Optional:
    ↪ uncomment for verbose skipping
                continue

            try:
                # 1. Load image
                img = io.imread(img_path)

                # 2. Convert to grayscale if it's a color image (handles RGB/
    ↪ RGBA)
                if img.ndim == 3 and img.shape[2] >= 3:

```

```

        img_gray = color.rgb2gray(img)
    else:
        # If it's already grayscale or single channel, use it as is.
        # Ensure it's float for subsequent operations if it wasn't
        ↪ already.

        img_gray = img.astype(float)

        # 3. Resize image to a consistent size (e.g., 100x100)
        # anti_aliasing=True is recommended for downsampling to prevent
        ↪ aliasing artifacts.

        img_resized = transform.resize(img_gray, img_size,
        ↪ anti_aliasing=True)

        # 4. Apply Histogram Equalization for contrast enhancement
        # This operation typically outputs float values in the range [0.
        ↪ 0, 1.0].

        img_eq = exposure.equalize_hist(img_resized)

        # Convert the float image (0.0-1.0) to uint8 (0-255) for common
        ↪ image formats like JPEG

        # Multiplying by 255 and casting to uint8 scales the pixel
        ↪ values appropriately.

        img_to_save = (img_eq * 255).astype(np.uint8)

        # 5. Define the full path for the processed image in the output
        ↪ directory

        output_img_path = output_folder_path / img_filename

        # 6. Save the processed image
        io.imwrite(output_img_path, img_to_save)
        print(f"Processed and saved: {output_img_path}")

    except Exception as e:
        # Catch any errors during image processing (e.g., corrupted
        ↪ files, unsupported formats)

        print(f"Error processing {img_path}: {e}")

input_data_directory = "data/images" # Example: "C:/Users/YourUser/MyDataset"
        ↪ or "/home/user/data/images"
output_results_directory = "processed_images"

print(f"Starting image preprocessing from: {os.path.
        ↪ abspath(input_data_directory)}")
print(f"Output will be saved to: {os.path.abspath(output_results_directory)}")

```

```
# Call the image preprocessing function
preprocess_images_skimage(input_data_directory, output_results_directory)

print("\nImage preprocessing task completed!")
```

Starting image preprocessing from: /Users/kishankunal/PycharmProjects/AppliedAI/RailwayTwitterSentimentAnalysis/data/images

Output will be saved to: /Users/kishankunal/PycharmProjects/AppliedAI/RailwayTwitterSentimentAnalysis/processed_images

Processed and saved: processed_images/0/Anger.jpg
Processed and saved: processed_images/0/Contempt.jpg
Processed and saved: processed_images/0/Disgust.jpg
Processed and saved: processed_images/0/Fear.jpg
Processed and saved: processed_images/0/Happy.jpg
Processed and saved: processed_images/0/Sad.jpg
Processed and saved: processed_images/0/Surprised.jpg
Processed and saved: processed_images/0/Neutral.jpg
Processed and saved: processed_images/1/Anger.jpg
Processed and saved: processed_images/1/Contempt.jpg
Processed and saved: processed_images/1/Disgust.jpg
Processed and saved: processed_images/1/Fear.jpg
Processed and saved: processed_images/1/Happy.jpg
Processed and saved: processed_images/1/Sad.jpg
Processed and saved: processed_images/1/Surprised.jpg
Processed and saved: processed_images/1/Neutral.jpg
Processed and saved: processed_images/2/Anger.jpg
Processed and saved: processed_images/2/Contempt.jpg
Processed and saved: processed_images/2/Disgust.jpg
Processed and saved: processed_images/2/Fear.jpg
Processed and saved: processed_images/2/Happy.jpg
Processed and saved: processed_images/2/Sad.jpg
Processed and saved: processed_images/2/Surprised.jpg
Processed and saved: processed_images/2/Neutral.jpg
Processed and saved: processed_images/3/Anger.jpg
Processed and saved: processed_images/3/Contempt.jpg
Processed and saved: processed_images/3/Disgust.jpg
Processed and saved: processed_images/3/Fear.jpg
Processed and saved: processed_images/3/Happy.jpg
Processed and saved: processed_images/3/Sad.jpg
Processed and saved: processed_images/3/Surprised.jpg
Processed and saved: processed_images/3/Neutral.jpg
Processed and saved: processed_images/4/Anger.jpg
Processed and saved: processed_images/4/Contempt.jpg
Processed and saved: processed_images/4/Disgust.jpg
Processed and saved: processed_images/4/Fear.jpg
Processed and saved: processed_images/4/Happy.jpg
Processed and saved: processed_images/4/Sad.jpg

Processed and saved: processed_images/4/Surprised.jpg
Processed and saved: processed_images/4/Neutral.jpg
Processed and saved: processed_images/5/Anger.jpg
Processed and saved: processed_images/5/Contempt.jpg
Processed and saved: processed_images/5/Disgust.jpg
Processed and saved: processed_images/5/Fear.jpg
Processed and saved: processed_images/5/Happy.jpg
Processed and saved: processed_images/5/Sad.jpg
Processed and saved: processed_images/5/Surprised.jpg
Processed and saved: processed_images/5/Neutral.jpg
Processed and saved: processed_images/6/Anger.jpg
Processed and saved: processed_images/6/Contempt.jpg
Processed and saved: processed_images/6/Disgust.jpg
Processed and saved: processed_images/6/Fear.jpg
Processed and saved: processed_images/6/Happy.jpg
Processed and saved: processed_images/6/Sad.jpg
Processed and saved: processed_images/6/Surprised.jpg
Processed and saved: processed_images/6/Neutral.jpg
Processed and saved: processed_images/7/Anger.jpg
Processed and saved: processed_images/7/Contempt.jpg
Processed and saved: processed_images/7/Disgust.jpg
Processed and saved: processed_images/7/Fear.jpg
Processed and saved: processed_images/7/Happy.jpg
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Processed and saved: processed_images/7/Surprised.jpg
Processed and saved: processed_images/7/Neutral.jpg
Processed and saved: processed_images/8/Anger.jpg
Processed and saved: processed_images/8/Contempt.jpg
Processed and saved: processed_images/8/Disgust.jpg
Processed and saved: processed_images/8/Fear.jpg
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Processed and saved: processed_images/8/Neutral.jpg
Processed and saved: processed_images/9/Anger.jpg
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Processed and saved: processed_images/9/Surprised.jpg
Processed and saved: processed_images/9/Neutral.jpg
Processed and saved: processed_images/10/Anger.jpg
Processed and saved: processed_images/10/Contempt.jpg
Processed and saved: processed_images/10/Disgust.jpg
Processed and saved: processed_images/10/Fear.jpg
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Processed and saved: processed_images/10/Surprised.jpg
Processed and saved: processed_images/10/Neutral.jpg
Processed and saved: processed_images/11/Anger.jpg
Processed and saved: processed_images/11/Contempt.jpg
Processed and saved: processed_images/11/Disgust.jpg
Processed and saved: processed_images/11/Fear.jpg
Processed and saved: processed_images/11/Happy.jpg
Processed and saved: processed_images/11/Sad.jpg
Processed and saved: processed_images/11/Surprised.jpg
Processed and saved: processed_images/11/Neutral.jpg
Processed and saved: processed_images/12/Anger.jpg
Processed and saved: processed_images/12/Contempt.jpg
Processed and saved: processed_images/12/Disgust.jpg
Processed and saved: processed_images/12/Fear.jpg
Processed and saved: processed_images/12/Happy.jpg
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Processed and saved: processed_images/15/Fear.jpg
Processed and saved: processed_images/15/Happy.jpg
Processed and saved: processed_images/15/Sad.jpg
Processed and saved: processed_images/15/Surprised.jpg
Processed and saved: processed_images/15/Neutral.jpg
Processed and saved: processed_images/16/Anger.jpg
Processed and saved: processed_images/16/Contempt.jpg
Processed and saved: processed_images/16/Disgust.jpg
Processed and saved: processed_images/16/Fear.jpg
Processed and saved: processed_images/16/Happy.jpg
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Processed and saved: processed_images/16/Surprised.jpg
Processed and saved: processed_images/16/Neutral.jpg
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Processed and saved: processed_images/17/Contempt.jpg
Processed and saved: processed_images/17/Disgust.jpg
Processed and saved: processed_images/17/Fear.jpg
Processed and saved: processed_images/17/Happy.jpg
Processed and saved: processed_images/17/Sad.jpg
Processed and saved: processed_images/17/Surprised.jpg
Processed and saved: processed_images/17/Neutral.jpg
Processed and saved: processed_images/18/Anger.jpg
Processed and saved: processed_images/18/Contempt.jpg
Processed and saved: processed_images/18/Disgust.jpg
Processed and saved: processed_images/18/Fear.jpg
Processed and saved: processed_images/18/Happy.jpg
Processed and saved: processed_images/18/Sad.jpg
Processed and saved: processed_images/18/Surprised.jpg
Processed and saved: processed_images/18/Neutral.jpg

Image preprocessing task completed!

[]: